

KinetiDiff: Docking-Guided Diffusion for De Novo ACVR1 Inhibitor Design in Fibrodysplasia Ossificans Progressiva

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Abstract—Fibrodysplasia Ossificans Progressiva (FOP) is a rare, progressive disorder caused by gain-of-function mutations in the ACVR1/ALK2 kinase, for which no curative treatment exists and the chemical space of potent, selective inhibitors remains largely unexplored. Current candidates are few, emerged from conventional medicinal chemistry, and cover only a narrow slice of tractable inhibitor space.

We present KinetiDiff, a structure-based generative framework that couples a Geometry-Complete Diffusion Model (GCDM) with real-time AutoDock Vina gradient guidance—injecting physics-based docking signals directly into the reverse diffusion loop rather than filtering molecules after generation. Applied to ACVR1 (PDB: 3MTF), the framework generates a large, chemically diverse pool of Lipinski-compliant inhibitor candidates whose predicted affinities broadly surpass the crystallographic reference. Systematic ablation across four guidance strategies shows that real-time physics-based guidance outperforms both learned surrogate guidance and unguided generation across affinity, drug-likeness, and synthetic accessibility; in silico ADMET profiling of the top candidates further indicates pharmacokinetic properties consistent with oral drug candidates.

The central aim is not to claim a drug but to demonstrate that physics-guided diffusion can systematically explore ACVR1 inhibitor space from scratch—producing a structurally diverse, computationally validated pool of leads and opening a structured path toward experimental follow-up via binding assays, analog synthesis, and molecular dynamics.

Index Terms—diffusion models, structure-based drug design, AutoDock Vina, ACVR1/ALK2, fibrodysplasia ossificans progressiva, gradient guidance, geometric deep learning

I. INTRODUCTION

Fibrodysplasia Ossificans Progressiva (FOP) is a rare autosomal-dominant disorder caused by gain-of-function mutations in *ACVR1*, encoding the ALK2 kinase. The R206H mutation renders ALK2 constitutively active and responsive to activin A, driving SMAD1/5/8 phosphorylation and progressive heterotopic ossification of soft tissues [1], [2]. Current inhibitors—saracatinib, zilurgesertib, palovarotene—show efficacy but carry off-target activity or incomplete mechanistic targeting [3], [4].

Generative models offer an alternative: rather than screening libraries, they design molecules de novo conditioned on target structure [5], [6], [7]. Diffusion models operating in 3D coordinate space with geometric inductive biases are particularly effective for structure-conditioned ligand generation [8], [9], [10], [11], [12], [13].

However, most diffusion-based SBDD pipelines *decouple* generation from binding evaluation: molecules are sampled un-

conditionally or with pocket-shape conditioning, then filtered post hoc. The diffusion model never sees the actual docking score during sampling, limiting steering toward high-affinity solutions. While recent work on classifier-guided diffusion addresses this [11], [12], no prior framework has integrated real-time AutoDock Vina scoring into denoising for a therapeutic target.

KinetiDiff injects AutoDock Vina gradients into the reverse diffusion process. We evaluate it against three alternatives—HNN-Denovo, multi-objective, and unguided generation—demonstrating superior performance across all metrics. Applied to ACVR1, it generates 9,997 valid molecules, with the best achieving -11.21 kcal/mol (20.9% over the crystallographic reference). Code is available at: <https://anonymous.4open.science/r/KinetiDiff-31EB>.

II. BACKGROUND

A. Denoising Diffusion Probabilistic Models

A DDPM defines a Markov chain corrupting data $\mathbf{x}_0$ over T timesteps. Using $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$, the forward marginal is:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad (1)$$

enabling direct sampling as $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$. A neural network ϵ_θ is trained to predict the noise, with the connection to the score function:

$$\epsilon_\theta(\mathbf{x}_t, t) \approx -\sqrt{1 - \bar{\alpha}_t} \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t). \quad (2)$$

This relationship enables guidance: modifying the score with an auxiliary gradient steers generation toward desired properties. We adopt a cosine noise schedule [14] with offset $s = 0.008$.

B. GCDM: SE(3)-Equivariant 3D Diffusion

The Geometry-Complete Diffusion Model (GCDM) [8] extends DDPMs to 3D molecules through SE(3)-equivariant message passing via the Geometry-Complete Perceptron (GCP). Hidden representations split into invariant *scalar* channels $\mathbf{s} \in \mathbb{R}^{d_s}$ and equivariant *vector* channels $\mathbf{v} \in \mathbb{R}^{3 \times d_v}$, ensuring that rotating the molecule rotates the predicted coordinates correspondingly.

For pocket-conditioned generation, GCDM defines the binding site neighborhood as:

$$\mathcal{N}_c = \{(\mathbf{v}_i, \mathbf{r}_i) \mid \|\mathbf{r}_i - \mathbf{r}_c\| \leq 10 \text{ \AA}\}, \quad (3)$$

where $\mathbf{r}_c = (24.87, -12.54, 38.40)$ Å is the ACVR1 binding site centroid (PDB: 3MTF). A pocket embedding $\mathbf{c}_t$ is formed by distance-weighted pooling of equivariant residue features and injected additively into each denoising layer.

III. METHODOLOGY

A. Framework Overview

KinetiDiff operates in three stages: (i) pocket-conditioned 3D generation via GCDM with GCPNet; (ii) real-time Vina gradient guidance during reverse diffusion; and (iii) multi-objective post-hoc ranking. Generation and guidance are unified in a single sampling loop (Figure 1).

B. Real-Time Vina Gradient Guidance

The central innovation is injecting AutoDock Vina [15] gradients directly into the denoising loop. At qualifying timesteps, we shift atomic coordinates toward higher binding affinity:

$$\mathbf{x}_{t-1} = \hat{\mathbf{x}}_0^{(t)} - \lambda_t \hat{\mathbf{g}}_t, \quad (4)$$

where $\hat{\mathbf{x}}_0^{(t)}$ is the GCDM’s predicted clean coordinates, λ_t is the timestep-dependent guidance scale, and $\hat{\mathbf{g}}_t$ is the normalized numerical Vina gradient.

a) *Numerical Gradient.*: Because Vina’s scoring function is not differentiable, we apply central finite differences:

$$\hat{\mathbf{g}}_t = \sum_{i=1}^N \sum_{d \in \{x,y,z\}} \frac{\mathcal{U}_{\text{Vina}}(\mathbf{x}_t + \varepsilon \mathbf{e}_{i,d}) - \mathcal{U}_{\text{Vina}}(\mathbf{x}_t - \varepsilon \mathbf{e}_{i,d})}{2\varepsilon} \mathbf{e}_{i,d}, \quad (5)$$

with step size $\varepsilon = 0.05$ Å. Each evaluation requires $6N + 1$ Vina scoring calls. Gradients are norm-clipped to 10.0 then normalized to unit length.

b) *Guidance Schedule.*: Strength is modulated by an exponential schedule with an activation indicator:

$$\lambda_t = [0.1 \cdot \exp(3(1 - \frac{t}{T}))]_{0.1}^{10.0} \cdot \mathbb{1}_{[t \leq 0.4T]}, \quad (6)$$

suppressing guidance during early, noisy timesteps (high t) where molecular geometry is incoherent, and activating it only once the structure is sufficiently coherent ($t \leq 0.4T$). Guidance is applied every 10 denoising steps within the active window $t \in [0, 0.4T]$. After each guided step, coordinates are re-centered ($\mathbf{x} \leftarrow \mathbf{x} - \frac{1}{N} \sum_i \mathbf{x}_i$) to maintain the zero center-of-mass constraint required by SE(3)-equivariant diffusion.

c) *Docking Configuration.*: All guidance-phase Vina evaluations use local docking mode (exhaustiveness 1) centered at $(24.87, -12.54, 38.40)$ Å with a $22 \times 22 \times 22$ Å search box and seed 42.

C. HNN-Denovo: Neural Proxy Guidance

As a computationally efficient alternative, we evaluate HNN-Denovo, a convolutional neural network trained on 15,324 ACVR1-family protein–ligand pairs from BindingDB [16]. The network uses parallel 1D convolutional branches (kernels $k \in \{3, 5, 7\}$, 256 channels) for protein sequences and ligand SMILES, predicting binding affinity (pKd). Test-set performance: PCC = 0.72, RMSE = 0.70 pKd.

For gradient guidance, a differentiable 3D proxy score is computed from atomic coordinates via contact counts ($\sigma((\tau - d_{ij}) \cdot \kappa)$, $\tau = 4.5$ Å) and inverse-distance potentials. This enables backpropagation without finite-difference approximation, yielding a $60\times$ speedup per guidance step. However, the proxy achieves only $r = 0.224$ correlation with Vina ground truth, limiting guidance precision (Section IV-B).

D. Multi-Objective Scoring

Generated molecules are ranked using a composite score:

$$S(\mathbf{x}) = 0.5 \cdot S_{\text{aff}} + 0.3 \cdot S_{\text{kin}} + 0.2 \cdot S_{\text{SA}}, \quad (7)$$

where S_{aff} is a Gaussian on pKd (target 7.5, $\sigma = 1.5$), S_{kin} rewards $k_{\text{off}} \in [0.1, 1.0] \text{ s}^{-1}$, and S_{SA} penalizes SA > 4.0. Pareto-optimal molecules provide diverse trade-off solutions.

E. Docking Validation Protocol

All candidates were uniformly redocked against ACVR1 (PDB: 3MTF) using AutoDock Vina 1.2.3: $22 \times 22 \times 22$ Å box centred on the crystallographic A3F ligand centroid $(-17.52, -13.58, 38.33)$ Å, exhaustiveness 8, seed 42. 3D conformers used RDKit ETKDGV3, energy-minimized with MMFF94, and converted to PDBQT via Meeko. The receptor was prepared from chain A only (no crystallographic waters or co-ligands) to ensure an unoccupied binding site. The crystallographic reference (A3F) redocked at -9.27 kcal/mol.

IV. RESULTS

A. Generation Statistics

Vina-Direct produced 9,997 valid molecules from 10,000 samples (99.97%). Multi-objective yielded 3,524 valid structures from 5,000 samples (70.5%), with lower validity from more aggressive pocket-conditioning. HNN-Denovo produced 1,500 molecules across three runs. An unguided baseline of 4,000 molecules serves as the reference distribution.

B. Ablation: Guidance Strategy Comparison

All 19,021 molecules were assessed on pKd, QED, SA, and LogP (Figure 2). Vina-Direct guidance dominates across all metrics: median pKd 6.71 vs. HNN-Denovo 6.29 (+0.42 log units) and Multi-Objective 5.52 (+1.19 log units). Critically, Vina-Direct molecules are also the easiest to synthesize (median SA 2.26 vs. 4.97 for HNN-Denovo), because physics-based Vina scoring rewards compact, well-organized geometries that correlate with synthetic tractability.

HNN-Denovo improves over the unguided baseline by 0.75 pKd units, confirming that even an approximate signal biases generation toward active molecules. However, the differentiable 3D proxy used to bridge HNN-Denovo’s SMILES domain into coordinate space achieves only $r = 0.224$ against Vina, explaining the SA penalty (median 4.97) from distance-based approximations that do not effectively enforce synthetic tractability.

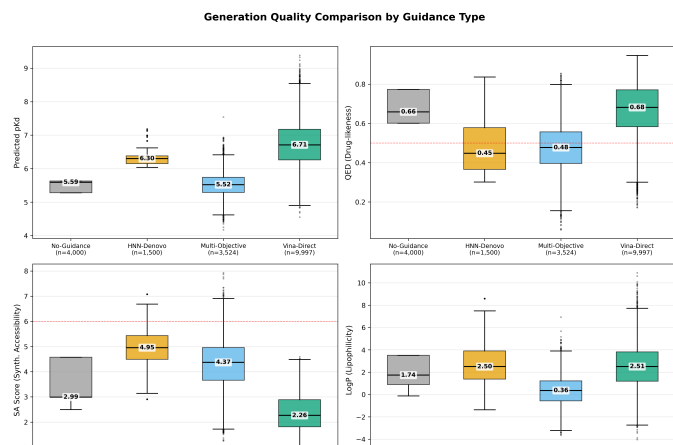


Fig. 2: Comparison of four guidance strategies across binding affinity (pKd), drug-likeness (QED), synthetic accessibility (SA), and lipophilicity (LogP). Vina-Direct (green) achieves the highest affinity, best drug-likeness, and lowest synthetic complexity.

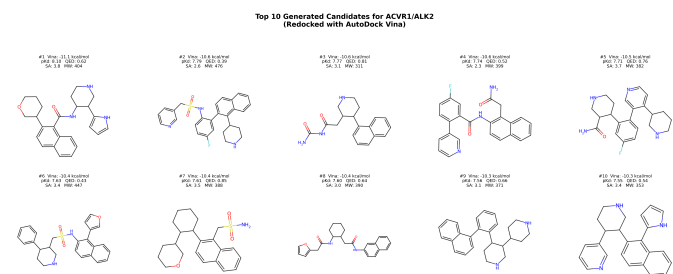


Fig. 3: 2D structures of the top 10 generated ACVR1 inhibitor candidates with docking scores and drug-likeness properties.

show predicted high gastrointestinal absorption, zero Lipinski rule-of-five violations, zero PAINS alerts, and a bioavailability score of 0.55 — properties broadly consistent with oral drug candidates. Molecular weights span 212–476 Da, consensus logP ranges from 0.06 to 4.74, and TPSA from 24 to 104 Å², all within established oral bioavailability thresholds (MW < 500 Da, logP < 5, TPSA < 140 Å²). Fourteen of 16 molecules are classified as *Soluble* or *Very soluble* by ESOL; the remaining two (Molecules 1 and 15) fall in the *Moderately soluble* range, which is generally acceptable for lead optimization. Nine candidates are predicted blood–brain–barrier permeable. For FOP—a systemic soft-tissue disorder—CNS penetration is not a primary therapeutic requirement, making BBB permeability a pharmacologically neutral attribute.

Seven molecules carry one or more Brenk structural alerts, which flag fragments with historical associations with metabolic or reactive liabilities. These alerts warrant attention during lead refinement but do not exclude compounds from further profiling. Synthetic accessibility scores range from 2.79 (Molecule 3, most tractable) to 5.42 (Molecule 8), with twelve of sixteen candidates scoring below 5.0 and ten below 4.5, a



Fig. 4: Property distributions by guidance type. Lipinski thresholds shown as red dashed lines. Vina-Direct molecules (green) cluster within drug-like space.

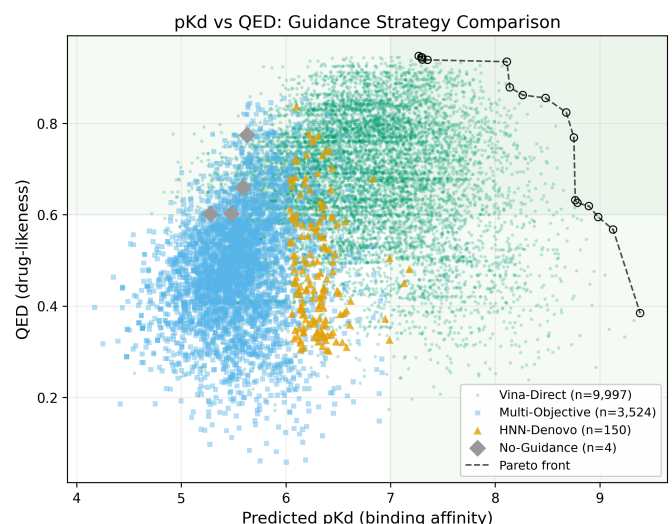


Fig. 5: pKd vs. QED across 19,021 molecules. The Pareto front (dashed) traces non-dominated trade-offs; Vina-Direct molecules (green) dominate the frontier.

common threshold for commercially feasible synthesis.

Taken together, the ADMET profiles indicate that the generated molecules are not merely computational artifacts optimized for docking score but carry pharmacokinetic characteristics compatible with downstream biological evaluation. Molecules 1–3, which correspond to the top three Vina-ranked leads, show particularly favorable profiles (TPSA 66–84 Å², cLogP 1.9–4.8, SA 2.8–4.1). These properties provide a principled basis for experimental prioritization, though they do not substitute for measured values.

V. DISCUSSION

a) Computational Findings.: (1) Real-time Vina gradient injection into GCDM produces ACVR1 inhibitor candidates whose predicted affinities exceed the crystallographic reference by up to 19.2%, while maintaining Lipinski compliance and synthetic tractability across structurally diverse scaffold

TABLE II: In silico ADMET profiles for the 16 top KinetiDiff candidates (SwissADME [17]). All 16 show High GI absorption, 0 Lipinski violations, 0 PAINS alerts, and bioavailability score 0.55. Molecules 1–3 correspond to the top three Vina-ranked leads in Table I. Brenk = number of structural alerts; SA = synthetic accessibility (1–10, lower is easier).

Mol.	MW (Da)	cLogP	TPSA ($\text{\AA}^2$)	BBB	Brenk	SA
1	403.5	3.14	66.2	Yes	0	4.09
2	475.6	4.74	79.5	No	0	3.60
3	311.4	1.95	84.2	No	0	2.79
4	327.9	2.76	52.5	Yes	0	4.56
5	307.5	1.43	27.7	No	0	3.97
6	212.3	1.87	24.1	Yes	0	3.61
7	293.4	2.18	62.1	Yes	2	5.12
8	363.5	1.62	103.5	No	0	5.42
9	262.4	1.60	48.5	Yes	2	4.44
10	323.5	1.63	73.4	No	2	4.40
11	327.4	2.34	71.6	Yes	1	4.93
12	352.6	3.09	55.7	Yes	0	5.22
13	302.4	0.06	85.2	No	0	4.26
14	270.4	2.62	70.4	Yes	2	4.38
15	291.5	3.58	32.3	Yes	0	3.93
16	250.4	2.77	40.5	Yes	1	4.54
<i>Oral bioavailability thresholds</i>		< 5.0	< 140	–	–	< 5.0

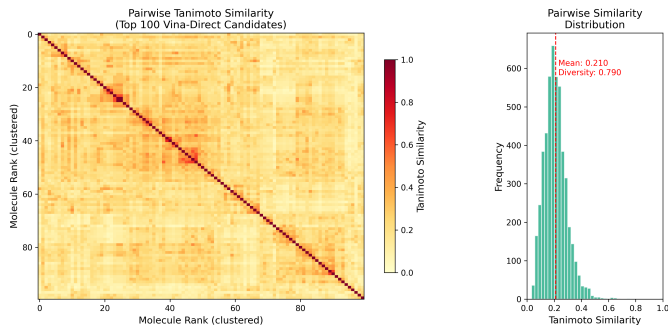


Fig. 6: Chemical diversity of the top 100 candidates. Left: pairwise Tanimoto heatmap (hierarchically clustered). Right: off-diagonal similarity distribution. Mean Tanimoto 0.210.

families. (2) Physics-based oracle guidance produces more favorable property profiles than learned proxies or unguided generation in this setting; HNN-Denovo’s underperformance traces directly to the low correlation ($r = 0.224$) between its SMILES-domain proxy and actual Vina gradients in 3D coordinate space, rather than to the underlying affinity model itself (PCC = 0.72). (3) In silico ADMET profiling shows that all 16 profiled candidates carry pharmacokinetic characteristics consistent with oral bioavailability, providing a principled first filter for experimental prioritization.

b) Vina-Direct vs. HNN-Denovo.: HNN-Denovo’s $60\times$ per-step speedup remains attractive for rapid, coarse-resolution exploration of novel targets where an approximate directional signal suffices. The lesson here is about domain mismatch: a SMILES-based model that performs well on sequence-level affinity prediction ($r = 0.72$) can fail to provide useful coordinate-space gradients when the bridge between domains is a poorly correlated proxy. For established, structurally well-characterized targets like ACVR1, the additional computational cost of direct Vina scoring is justified; for novel targets lacking validated structures, HNN-Denovo or analogous fast

surrogates may still serve as a productive first pass.

c) Limitations.: All binding evaluations use rigid-receptor Vina docking, which does not capture receptor flexibility, induced-fit conformational changes, or explicit solvation—known limitations for kinase targets where the DFG-loop and activation-loop conformations differ between WT and gain-of-function mutants. ADMET profiling is entirely computational (SwissADME); no experimental pharmacokinetic measurements have been performed. No kinase selectivity panel was run; cross-reactivity against related BMP receptor kinases (BMPRI1A, BMPRI1B) is unknown. Vina scores are proxies for binding affinity, not measured K_d or IC_{50} values. Dissociation kinetics are estimated from an empirical pK_d – k_{off} correlation. Critically, *no candidate has been synthesized or tested biologically*; all reported properties are computational predictions.

d) Paths to Experimental Validation.: The goal of this work is to demonstrate that physics-guided diffusion can populate FOP inhibitor space with pharmacokinetically plausible, structurally diverse lead candidates—not to claim a drug. Realizing the therapeutic potential of these candidates requires three complementary steps. (i) *Binding assays*: biochemical ACVR1 kinase inhibition assays (IC_{50}) and cell-based reporter assays for SMAD1/5/8 phosphorylation in FOP patient-derived cell lines would provide the first experimental evidence of biological activity, testing whether the predicted affinity translates to measured potency. (ii) *Analog synthesis and structure–activity relationships*: the scaffold families identified—urea-linked naphthyl-piperidines (Molecule 3), sulfonamide fluorobiphenyl-piperidines (Molecule 2), dual fluoropyridyl-piperidines (Molecule 5)—are tractable starting points for focused analog libraries that can delineate which pharmacophoric features drive activity and guide iterative lead optimization. (iii) *Molecular dynamics*: atomistic MD simulations of top leads against both WT and R206H ACVR1 would assess whether the constitutively-active mutant conformation selectively stabilizes inhibitor contacts, addressing

the FOP-specific question of mutant-over-WT selectivity that rigid docking cannot answer. Together, these steps define the experimental roadmap from computational generation to validated biological activity.

VI. CONCLUSION

KinetiDiff explores the application of physics-guided geometric diffusion to de novo ACVR1 inhibitor design for Fibrodysplasia Ossificans Progressiva. The framework generates chemically diverse, drug-like candidates with predicted binding affinities above the crystallographic reference and in silico ADMET profiles consistent with oral bioavailability. The central claim is not that any specific candidate constitutes a clinical lead, but that generative AI with real-time physics-based guidance can systematically map inhibitor space for a rare-disease target—producing a structurally diverse, pharmacokinetically filtered pool of starting points for experimental follow-up.

This work is a step in the computational phase of the drug discovery pipeline, not the conclusion of it. The most important remaining work is biological: synthesis of top candidates, kinase inhibition assays, cellular validation, and structure-activity guided analog optimization. Molecular dynamics simulations would further address the FOP-specific question of WT vs. R206H selectivity that rigid docking cannot resolve. We hope this framework illustrates the potential of physics-guided diffusion as a practical computational tool for rare diseases—FOP among them—where the therapeutic landscape remains limited and the patient need is acute.

REFERENCES

- [1] F. S. Kaplan, M. Le Merrer, D. L. Glaser, R. J. Pignolo, R. E. Goldsby, J. A. Kitterman, J. C. Groppe, and E. M. Shore, “Fibrodysplasia ossificans progressiva,” *Best Practice & Research Clinical Rheumatology*, vol. 22, no. 1, pp. 191–205, 2008.
- [2] D. Srinivasan, M. Arostegui, E. J. Goebel, K. N. Hart, S. Aykul, J. B. Lees-Shepard, V. Idone, S. J. Hatsell, and A. N. Economides, “How activin a became a therapeutic target in fibrodysplasia ossificans progressiva,” *Biomolecules*, vol. 14, no. 1, p. 101, 2024.
- [3] E. Williams, J. Bagarova, G. Kerr, D.-D. Xia, E. S. Place, D. Dey, Y. Shen, G. A. Bocobo, A. H. Mohedas, X. Huang, P. E. Sanderson, A. Lee, W. Zheng, A. N. Economides, J. C. Smith, P. B. Yu, and A. N. Bullock, “Saracatinib is an efficacious clinical candidate for fibrodysplasia ossificans progressiva,” *JCI Insight*, vol. 6, no. 7, p. e95042, 2021.
- [4] T. Ullrich, S. Guth, L. Arista, S. Weiler, N. Stiefl, S. Teixeira-Fouchard, C. Dekker, A. Hinniger, V. Head, M. Kneissel, and I. Kramer, “Discovery and characterization of zilurgesertib, a potent and selective inhibitor of activin receptor-like kinase-2 (alk2) for the treatment of fibrodysplasia ossificans progressiva,” *ACS Medicinal Chemistry Letters*, 2025.
- [5] A. Zhavoronkov, Y. A. Ivanenkov, A. Aliper, M. S. Veselov, V. A. Aladinskiy, A. V. Aladinskaya, V. A. Terentiev, D. A. Polykovskiy, M. D. Kuznetsov, A. Asadulaev, Y. Volkov, A. Zholus, R. R. Shayakhmetov, A. Zhebrak, L. I. Minaeva, B. A. Zagribelnyy, L. H. Lee, R. Soll, D. Madge, L. Xing, T. Guo, and A. Aspuru-Guzik, “Deep learning enables rapid identification of potent ddr1 kinase inhibitors,” *Nature Biotechnology*, vol. 37, no. 9, pp. 1038–1040, 2019.
- [6] Y. Li, L. Zhang, Y. Wang, J. Zou, R. Yang, X. Luo, C. Wu, W. Yang, C. Tian, H. Xu, F. Wang, X. Yang, L. Li, and S. Yang, “Generative deep learning enables the discovery of a potent and selective ripk1 inhibitor,” *Nature Communications*, vol. 13, no. 1, p. 6891, 2022.
- [7] U. Das, “Generative ai for drug discovery and protein design: the next frontier in ai-driven molecular science,” *Medicine in Drug Discovery*, vol. 27, p. 100213, 2025.

- [8] A. Morehead and J. Cheng, “Geometry-complete diffusion for 3d molecule generation and optimization,” *Communications Chemistry*, vol. 7, no. 1, p. 150, 2024.
- [9] X. Zheng, Y. Ren, T. Lai, X. Chen, L. P. Song, Q. Zhao, X. Wu, D. Bhowmik, and H. A. Carlson, “Apo2mol: 3d molecule generation via dynamic pocket-aware diffusion models,” *arXiv preprint arXiv:2404.08777*, 2024.
- [10] Q. Zhang and G. De Fabritiis, “Geometry-complete latent diffusion model for 3d molecule generation,” *Bioinformatics*, vol. 39, no. 12, p. btac781, 2025.
- [11] Y. Jian, C. Wu, D. Reidenbach, and A. S. Krishnapriyan, “General binding affinity guidance for diffusion models in structure-based drug design,” *arXiv preprint arXiv:2406.16821*, 2024.
- [12] Q. Hu, C. Sun, H. He, J. Xu, D. Liu, W. Zhang, S. Shi, K. Zhang, and H. Li, “Target-aware 3d molecular generation based on guided equivariant diffusion,” *Nature Communications*, 2025.
- [13] J. A. Weller, R. Rohs, B. A. Chapman, M. Meyer, M. De Montigny, L. Kuo, J. L. Wichman, M. D. Thomas, M. O’Meara, J. McPhail, L. Cheng, and A. Aspuru-Guzik, “Structure-based drug design with a deep hierarchical generative model,” *Journal of Chemical Information and Modeling*, vol. 64, no. 16, pp. 6450–6463, 2024.
- [14] A. Nichol and P. Dhariwal, “Improved denoising diffusion probabilistic models,” in *International Conference on Machine Learning*, 2021, pp. 8162–8171.
- [15] O. Trott and A. J. Olson, “Autodock vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading,” *Journal of Computational Chemistry*, vol. 31, no. 2, pp. 455–461, 2010.
- [16] T. Liu, Y. Lin, X. Wen, R. N. Jorissen, and M. K. Gilson, “Bindingdb: a web-accessible database of experimentally determined protein–ligand binding affinities,” *Nucleic Acids Research*, vol. 35, no. Database issue, pp. D198–D201, 2007.
- [17] A. Daina, O. Michielin, and V. Zoete, “SwissADME: a free web tool to evaluate pharmacokinetics, drug-likeness and medicinal chemistry friendliness of small molecules,” *Scientific Reports*, vol. 7, p. 42717, 2017.
- [18] Schrödinger, LLC, “The PyMOL molecular graphics system, version 3.1,” 2024. [Online]. Available: <https://pymol.org>
- [19] BioRender, “BioRender — science illustration tool,” 2026, accessed March 7, 2026. [Online]. Available: <https://biorender.com>

APPENDIX A

HYPERPARAMETERS

Table III summarizes the key GCDM architecture and Vina guidance parameters used in all experiments.

APPENDIX B

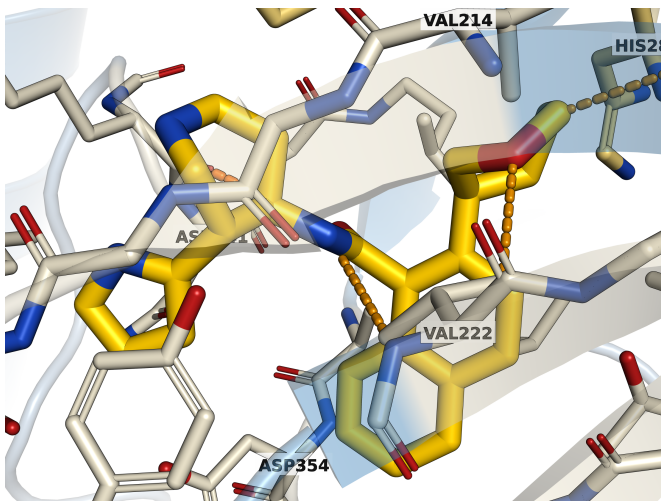
TOP LEAD DOCKING POSES

Figure 7 shows the two highest-ranked generated molecules docked in the ACVR1 ATP-binding pocket (PDB: 3MTF, chain A). Structures were rendered in PyMOL [18]: protein backbone as cartoon (light blue), pocket residues and ligand as sticks (ligand carbons in gold), and key heavy-atom contacts (≤ 3.6 Å) as orange dashed lines.

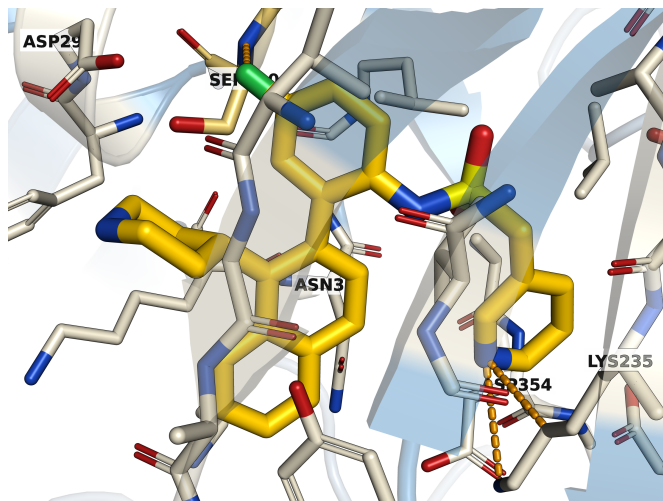
APPENDIX C

ACVR1/ALK2 SIGNALING PATHWAY

Figure 8 illustrates the molecular basis of heterotopic ossification in FOP and the rationale for kinase inhibition as a therapeutic strategy.



(a) Rank 1 (Vina: -11.21 kcal/mol, MW 403.5 Da, QED 0.618). Key contacts: TYR285, HIS286 (hinge), VAL222, VAL214, LYS235, ASN341, ASP354 (DFG).



(b) Rank 2 (Vina: -10.21 kcal/mol, MW 475.6 Da, QED 0.389). Key contacts: GLY289, SER290, LYS235, ASP293, ASN341, ASP354 (DFG). Sulfonamide NH donates to LYS235.

Fig. 7: Top two KinetiDiff candidates docked in the ACVR1 ATP-binding pocket (PDB: 3MTF). Orange dashed lines mark key pocket contacts (≤ 3.6 Å). Hinge residues 285–290 highlighted in gold.

TABLE III: GCDM and Vina guidance hyperparameters.

Parameter	Value
<i>GCPNet architecture</i>	
GCP interaction layers	6
Scalar hidden dim (d_s)	256
Vector hidden dim (d_v)	32
Edge RBF kernels	50 ($0-10$ Å, $\gamma = 0.5$ Å $^{-1}$)
Ligand atom types	9 (C, N, O, F, P, S, Cl, Br, I)
Pocket residue types	20 (standard amino acids)
Diffusion timesteps T	1000
DDIM inference steps	50 (stride 20, $\eta = 0$)
<i>Vina gradient guidance</i>	
Finite-difference ϵ	0.05 Å
Guidance active window	$t \in [0, 0.4T]$ (final 40%)
Guidance interval	Every 10 steps
λ clipping range	[0.1, 10.0]
Gradient norm clip	10.0
Docking mode (guidance)	-local_only, exhaustiveness 1
Docking mode (validation)	exhaustiveness 8
Box size	$22 \times 22 \times 22$ Å
Box center (guidance)	(24.87, -12.54, 38.40) Å †
Box center (validation)	(-17.52, -13.58, 38.33) Å
Random seed	42

† Guidance coordinates are in GCDM’s preprocessed pocket-centred frame (receptor PDB with polar-H addition); validation coordinates are the crystallographic A3F centroid in the raw RCSB coordinate frame. Both boxes target the same ATP-binding cleft of ACVR1 (PDB: 3MTF).

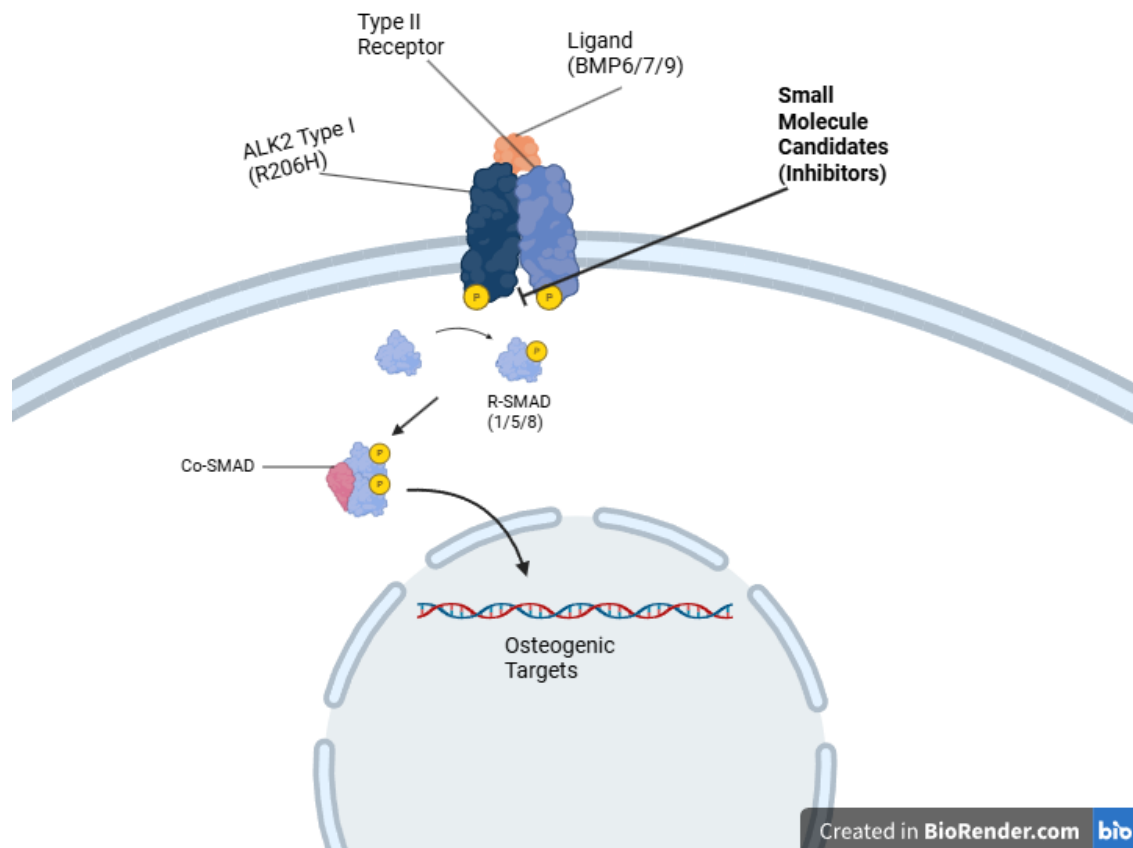


Fig. 8: ACVR1/ALK2 signaling cascade underlying heterotopic ossification in FOP [19]. The R206H gain-of-function mutation renders the ALK2 Type I receptor constitutively active upon BMP6/7/9 binding via the Type II receptor, driving phosphorylation of R-SMAD(1/5/8), heterodimerisation with Co-SMAD, and downstream transcriptional activation of osteogenic target genes. Small-molecule candidates generated by KinetiDiff target the ATP-binding cleft to suppress this pathological cascade.